

Computational structural dynamics

Project report



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PROJECT REPORT:

By using craig Bampton (CB) we got first 10 natural frequencies here used 15 eigen vector and size of matrix is [146 *146]

Sr.NO	ACTUAL (ω^2)	CB (ω^2)	%ERROR(CB)
1	0	0	0
2	0	0	0
3	0	0	0
4	19.5	19.5	0
5	22.1	22.1	0
6	147.3	146.5	-0.543
7	557.2	558.7	0.269
8	704.7	704.5	-0.028
9	1049.7	1050	0.02857
10	1173.2	1154.7	-1.5768

By using characteristic constraint mode (cc) we got first 10 natural frequencies here 15 eigen vector used and size of matrix is [75 *75] size of matrix gets reduce by using cc mode

Sr.NO	ACTUAL (ω^2)	CC (ω^2)	%ERROR(CC)
1	0	0	0
2	0	0	0
3	0	0	0
4	19.5	19.5	0
5	22.1	22.1	0
6	147.3	148.1	0.543
7	557.2	559.6	0.431
8	704.7	710.7	0.937
9	1049.7	1053.1	0.323
10	1173.2	1183.8	0.903

Conclusion: By using Characteristic constrain mode size of matrix reduce and it's seen that for lower frequencies up to 5th frequencies no error and above 5th frequencies there is error but less than 1 % percent.

```
clc; clear all ; close all ;
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```
load SubStructure1_Matlab.mat;
Tot_n1=1342; Int_n1=1256;
Bound_n1=86; N1=15;
kii=K1(1:Int_n1,1:Int_n1);
kib=K1(1:Int_n1,Int_n1+1:Tot_n1);
mii=M1(1:Int_n1,1:Int_n1); phi_c=-kii\kib;
[v1,d1]=eigs(kii,mii,N1,'sm'); rcb1=[v1
phi_c;zeros(Bound_n1,15) eye(Bound_n1)];
mr1=rcb1'*M1*rcb1; kr1=rcb1'*K1*rcb1;
```

```
% substructre 2 load
SubStructure2_Matlab.mat;
Tot_n2=462; Int_n2=440;
Bound_n2=22; N2=15;
kii=K2(1:Int_n2,1:Int_n2);
kib=K2(1:Int_n2,Int_n2+1:Tot_n2);
mii=M2(1:Int_n2,1:Int_n2); phi_c=-kii\kib;
[v2,d2]=eigs(kii,mii,N2,'sm'); rcb2=[v2
phi_c;zeros(Bound_n2,15) eye(Bound_n2)];
mr2=rcb2'*M2*rcb2; kr2=rcb2'*K2*rcb2;
```

```
%substructre 3 load
SubStructure3_Matlab.mat;
Tot_n3=882; Int_n3=840;
Bound_n3=42; N3=15;
kii=K3(1:Int_n3,1:Int_n3);
kib=K3(1:Int_n3,Int_n3+1:Tot_n3);
mii=M3(1:Int_n3,1:Int_n3); phi_c=-kii\kib;
[v3,d3]=eigs(kii,mii,N3,'sm'); rcb3=[v3
phi_c;zeros(Bound_n3,15) eye(Bound_n3)];
mr3=rcb3'*M3*rcb3; kr3=rcb3'*K3*rcb3;
```

```
%substructre 4 load
SubStructure4_Matlab.mat;
Tot_n4=462; Int_n4=440;
Bound_n4=22; N4=15;
kii=K4(1:Int_n4,1:Int_n4);
kib=K4(1:Int_n4,Int_n4+1:Tot_n4);
mii=M4(1:Int_n4,1:Int_n4); phi_c=-kii\kib;
[v4,d4]=eigs(kii,mii,N4,'sm'); rcb4=[v4
phi_c;zeros(Bound_n4,15) eye(Bound_n4)];
mr4=rcb4'*M4*rcb4; kr4=rcb4'*K4*rcb4; %
ASSEMBLY
r1=size(mr1);r2=size(mr2);r3=size(mr3);r4=size(mr4);
a=r1(1)+r2(1); b=r1(1)+r2(1)+r3(1);
c=r1(1)+r2(1)+r3(1)+r4(1);
```

```
M_assmeled=zeros(c);
M_assmeled(1:r1(1),1:r1(1))=mr1;
M_assmeled(r1(1)+1:a,r1(1)+1:a)=mr2;
M_assmeled(a+1:b,a+1:b)=mr3;
M_assmeled(b+1:c,b+1:c)=mr4;
```

```
K_assmeled=zeros(c);
K_assmeled(1:r1(1),1:r1(1))=kr1;
```

```

K_assmeled(r1(1)+1:a,r1(1)+1:a)=kr2;
K_assmeled(a+1:b,a+1:b)=kr3;
K_assmeled(b+1:c,b+1:c)=kr4;

```

```

%connectvity
T= zeros(c,c-86);
T(1:N1,1:N1)=eye(N1);
a=N1;b=N1+N2+N3+N4;
T(a+1:a+86,b+1:b+86)=eye(86);
a=a+86;b=N1;
T(a+1:a+N2,b+1:b+N2)=eye(N2);
a=a+N2;b=N1+N2+N3+N4;
T(a+1:a+22,b+1:b+22)=eye(22);
a=a+22;b=N1+N2;
T(a+1:a+N3,b+1:b+N3)=eye(N3);
a=a+N3;b=N1+N2+N3+N4+22;
T(a+1:a+42,b+1:b+42)=eye(42);
a=a+42;b=N1+N2+N3;
T(a+1:a+N4,b+1:b+N4)=eye(N4);
a=a+N4;b=N1+N2+N3+N4+22+42;
T(a+1:a+22,b+1:b+22)=eye(22);

```

```

% mfinal k final mf=T'*M_assmeled*T;
kf=T'*K_assmeled*T;

```

```

[vf,df]=eigs(kf,mf,10,'sm');

```

```

% CC Mode
a=N1+N2+N3+N4; N=15;
Kc=kf(a+1:a+86,a+1:a+86);
Mc=mf(a+1:a+86,a+1:a+86);
[vc,dc]=eigs(Kc,Mc,N,'sm');
Tc=zeros(a+86,a+N);
Tc(1:a,1:a)=eye(a);
Tc(a+1:a+86,a+1:a+N)=real(vc);
Mcc=Tc'*mf*Tc;
Kcc=Tc'*kf*Tc;
[vcc,dcc]=eigs(Kcc,Mcc,10,'sm');

```

